

What is RAG?

Retrieval-Augmented Generation (RAG) is a technique that combines document retrieval with a large language model. Instead of answering only from model weights, the system first searches a knowledge base, then generates an answer grounded in the retrieved passages.

A typical RAG system stores text chunks in a vector database. At query time it embeds the user question, finds similar chunks, and passes those chunks to the LLM as context. This reduces hallucinations on private or recent documents that the model was never trained on.

Basic metadata such as page number and document name should travel with each chunk so answers can cite sources.

The RAG pipeline

The core RAG pipeline has five steps. First, load PDFs and extract page text. Second, split pages into overlapping chunks so related sentences stay together. Third, embed each chunk with a sentence transformer such as BGE or E5. Fourth, upsert vectors into Qdrant. Fifth, at query time retrieve the top-k chunks and generate an answer.

Chunk size and overlap matter. Chunks around 800 characters with 150 characters of overlap are a practical starting point for technical PDFs. Too-small chunks lose context. Too-large chunks dilute similarity search.

LangGraph can wire retrieve and generate as two linear nodes in phase 1. Later phases add hybrid search, reranking, pruning, and retry edges.

Advanced RAG techniques

Advanced RAG improves retrieval quality before the LLM sees context. Hybrid search fuses dense vector results with BM25 keyword matches using Reciprocal Rank Fusion (RRF). A cross-encoder reranker then scores query-chunk pairs more accurately than bi-encoder similarity alone.

Context pruning drops low-score, duplicate, or token-heavy passages so the prompt stays small. Metadata filters restrict search to a section, document, or page range. Query rewrite retries retrieval when the first context is weak.

Prompt caching keeps a stable instruction prefix so the provider can reuse the KV cache. Observability with LangSmith records node spans for retrieve, rerank, prune, and generate.